

# K2-30b

R-2400

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_K2-30\_210109 · created 2026-08-21T00:28:23+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

## Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2459223.6914 +/- 0.0088 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.158 +/- 0.047                 |
| Transit depth (Rp/Rs)^2                           | 2.5 +/- 1.48 [%]                |
| Orbital Inclination (inc)                         | 87.19 +/- 2.35                  |
| Airmass coefficient 1 (a1)                        | 0.29 +/- 0.16                   |
| Airmass coefficient 2 (a2)                        | 1.2 +/- 0.74                    |
| Scatter in the residuals of the lightcurve fit is | 68.6 %                          |
| Best Comparison Star                              | #1 - [559, 144]                 |
| Optimal Aperture                                  | 5.43                            |
| Optimal Annulus                                   | 10.67                           |
| Transit Duration (day)                            | 0.082 +/- 0.035                 |

## Accuracy vs published ephemeris (ExoClock/NEA)

|                                      |                                                        |
|--------------------------------------|--------------------------------------------------------|
| Rp/R $\blacksquare$ fit vs published | 0.158 $\pm$ 0.047 vs 0.1266 (deviation 0.67 $\sigma$ ) |
| Duration fit vs published            | 0.082 d vs 0.0971 d (deviation 0.43 $\sigma$ )         |
| Mid-time O-C                         | -19.1 min                                              |
| Depth SNR                            | 3.4                                                    |
| Residual scatter                     | 68.6 %                                                 |

## Light curve

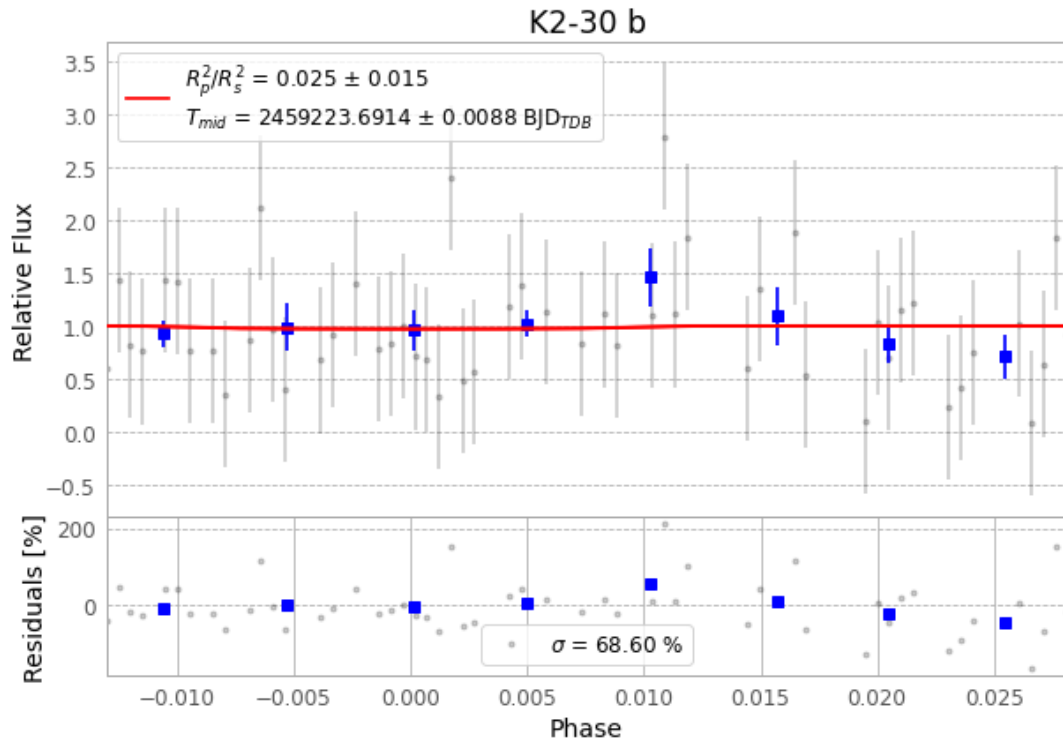


Figure 1 — FinalLightCurve\_K2-30 b\_2021-01-08.png (SHA-256 9fbaacef4cc0d92a...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [524, 133], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 643e6a4c77a64192...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit a4253ce

## Data provenance

81 FITS frames (53 MB), K2-30210109031214.FITS → K2-30210109071812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-K2-30-210109.log (e3eeae49bcf5...), FinalLightCurve\_K2-30 b\_2021-01-08.png (9fbaacef4cc0...), FinalLightCurve\_K2-30 b\_2021-01-08.csv (df12fed87a11...)

## Compute resources

Wall time 230.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

---

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-21 00:28Z.